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/*****
*   COMPANY                      : TAKEDA
*   PROTOCOL/STUDY/PROJECT      : ShortCourse
*   AUTHOR                      : OS Adjusted Analysis Working Group
*   DATE                        : 08Sep2025
*   PROGRAM NAME                : macro_adj_pvalue.sas
*   LOCATION                    : /.../.../...
*   VERSION/ PLATFORM          : SAS V9.4 / LINUX
*   PURPOSE                     : macro to calculate z-score and p-value from weighted Log-rank test
*   MACROS CALLED               :
*   INPUT                       :
*   OUTPUT                      :
*   PARAMETERS (Optional)      : INMAIN:      Name of final weighted analysis data
*                               : METHOD:      Weighted method
*                               :
*   NOTES (Optional)           : ASA Traveling Course, September to October 2025
*
*****/

%macro macro_adj_pvalue(inds=,method=);
ods exclude all;
*** from stablized weight output;

data ads0;
    set &inds;
run;

*** reorganize data for proc phreg;

proc sort data=ads0 out=ads1;
    by usubjid;
run;

proc univariate data=ads1 noprint;
    var analdy;
    output out=max_analdy max=maxday;
run;

data max_analdy;
    set max_analdy;
    maxday=round(maxday);

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run;

data _null_;
    set max_analdy;
    call symput('maxd', put(maxday,best.));
run;

data ads1;
    set ads1;
    by usubjid;

    if last.usubjid;
    keep usubjid;
run;

data ads1;
    set ads1;
    by usubjid;
    array a{&maxd.};

    do i=1 to &maxd.;
        a(i)=i;
    end;
    drop i;
run;

*** first, set correct count for atRisk;

proc transpose data=ads1 out=ads1;
    by usubjid;
run;

data ads1;
    set ads1;
    drop _name_;
    rename col1=analdy;
run;

proc sort data=ads1;
    by usubjid descending analdy;
run;

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proc sort data=ads0 out=ads2;
    by usubjid descending analdy;
run;

data ads2;
    merge ads1 ads2;
    by usubjid descending analdy;
    day=round(1*analdy);
    keep usubjid die stabw trt altrx day;
run;

data ads2;
    set ads2;
    by usubjid descending day;
    retain wt rx alrx;

    if first.usubjid and stabw=. then
        wt=-1;
    else
        do;

            if stabw ne . then
                wt=stabw;

            if trt ne . then
                rx=trt;

            if altrx ne . then
                alrx=altrx;
        end;
run;

data ads2;
    set ads2;
    by usubjid descending day;

    if wt=-1 then
        delete;

    if die=. then
        die=0;
    wtdie=die*wt;

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run;

proc sort data=ads2;
    by usubjid day die;
run;

data ads2;
    set ads2;
    by usubjid day die;

    if last.day;
run;

proc sort data=ads2;
    by rx day;
run;

proc univariate data=ads2 noprint;
    var wt die wtdie;
    by rx day;
    output out=main_temp n=atRisk temp1 temp2 sum=sumWgts totDead wgtDead;
run;

data main_plt;
    set main_temp;
    drop temp1 temp2;
run;

proc sort data=main_plt;
    by rx day;
run;

data main_plt;
    set main_plt;
    by rx day;

    if last.day;
run;

*** get normlized weight;

proc sort data=ads2;

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        by rx day usubjid;
run;

data nsw;
    merge ads2(keep=rx alrx day usubjid die wt) main_plt(keep=rx day atRisk
        sumWgts totDead wgtDead);
    by rx day;
    nsw=(wt*atRisk)/sumWgts;
run;

*****;
*** get G statistic;
*****;

data g0;
    set main_plt;

    if rx=0;
    rename SumWgts=WtRisk0 wgtDead=wtDie0;
    keep day sumWgts wgtDead;
run;

data g1;
    set main_plt;

    if rx=1;
    rename SumWgts=WtRisk1 wgtDead=wtDie1;
    keep day sumWgts wgtDead;
run;

*** construction riskset only when there is event, otherwise, no information;
*** without delete it, same result;

data gStat;
    merge g0 g1;
    by day;
    id=1;

    if wtRisk0=. or wtRisk1=. then
        delete;

    if wtDie0=0 and wtDie1=0 then

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        delete;
run;

data gStat1;
    set gStat;
    by id day;
    Gstat=(wtDie1-WtRisk1*sum(wtDie0, wtDie1)/sum(WtRisk0, WtRisk1));
run;

proc univariate data=gStat1 noprint;
    var Gstat;
    output out=gStat2 sum=Gstat1;
run;

data gStat2;
    set gStat2;
    id=1;
run;

*****;
*** get variance of Gstat;
*****;
*** inner loop, for subjects in each group, only loop on weights, as other variables are by j, not i;
*** per the paper, normalized weight need to be used;

data vg0;
    set nsw;

    if rx=0;
    wt0=nsw**2;
    keep day wt0;
run;

proc univariate data=vg0 noprint;
    var wt0;
    output out=wt1 sum=sumwt0;
    by day;
run;

data vgl;
    set nsw;

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        if rx=1;
        wt1=nsw**2;
        keep day wt1;
run;

proc univariate data=vgl noprint;
    var wt1;
    output out=wt2 sum=sumwt1;
    by day;
run;

data wt;
    merge wt1 wt2 g0 g1;
    by day;

    if sumwt0=. or sumwt1=. then
        delete;
    WtRisk=sum(WtRisk0, WtRisk1);
    vv=sum((WtRisk0/WtRisk)**2*sumwt0, (WtRisk1/WtRisk)**2*sumwt1);
run;

data wt;
    set wt;
    by day;
    id=1;
run;

*** for outloop;

data gv0;
    set main_plt;

    if rx=0;
    rename atRisk=Risk0 TotDead=Die0;
    id=1;
    keep day atRisk TotDead id;
run;

data gv1;
    set main_plt;

    if rx=1;

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        rename atRisk=Risk1 TotDead=Die1;
        id=1;
        keep id day atRisk TotDead;
run;

data getV;
    merge gv0 gv1 wt;
    by id day;

    if Risk0=. or Risk1=. then
        delete;
    atRisk=sum(Risk0, Risk1);
    die=sum(die0, die1);
run;

*** construction riskset only when there is event, otherwise, no information;
*** without delete it, same result;

data getV1;
    set getV;
    by id day;

    if die=0 then
        delete;
    sel=((die*(atRisk-die))/(atRisk*(atRisk-1)))*vv;
run;

proc univariate data=getV1 noprint;
    var sel;
    output out=getV2 sum=var;
run;

data getV2;
    set getV2;
    id=1;
run;

data stat_uspv_&method;
    merge getV2 gStat2;
    by id;
    z=Gstat1/sqrt(var);
    pvalue=1-probchi(z**2, 1);

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run;

ods exclude none;

proc print data=stat_uspv_&method;
    var z pvalue;
    title "z-score and p-value from weighted Log-rank test";
    title2 "Analysis is not stratified, adjusted by &Method";
run;
%mend macro_adj_pvalue;
```